Hypothesis Testing

Section 1: Null & Alternative Hypotheses, Type I & Type II Errors

1 Define a Type I error in hypothesis testing.

a. Rejecting H_0 when H_0 is actually true

b. Failing to reject H_0 when H_0 is false

c. Setting the significance level too high

d. Accepting H_1 when it is false

2 Define a Type II error in hypothesis testing.

a. Failing to reject H_0 when H_0 is false

b. Rejecting H_0 when H_0 is true

c. Computing an incorrect test statistic

d. Rejecting H_1 when H_0 is true

3 A machine produces metal pins with mean length 5.0 cm. A technician suspects the machine is out of adjustment and producing pins of different mean length. What are the appropriate hypotheses?

a. $H_0 : \mu = 5.0, H_1 : \mu \neq 5.0$

b. $H_0 : \mu = 5.0, H_1 : \mu > 5.0$

c. $H_0 : \mu \neq 5.0, H_1 : \mu = 5.0$

d. $H_0 : \mu = 5.0, H_1 : \mu < 5.0$

Section 2: Testing Population Mean with Normal Distribution

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4 A test of $H_0 : \mu = 80$ against $H_1 : \mu > 80$ at the 5% level is conducted. A sample of $n = 36$ from $N(\mu, 144)$ gives $\bar{x} = 83.5$. What is the value of the test statistic z?

a. $z = 1.75$

b. $z = 1.645$

c. $z = 2.00$

d. $z = 0.875$

5 For the test in the previous question ($z = 1.75$), what is the test conclusion at the 5% significance level (critical value 1.645)?

a. Reject H_0; there is sufficient evidence that the mean is greater than 80

b. Fail to reject H_0; insufficient evidence

c. Accept H_0; the difference is due to chance

d. Test is inconclusive

6 What is the p-value for the test statistic $z = 1.75$ in a one-tailed test?

a. 0.0401

b. 0.0802

c. 0.9599

d. 0.0250

Section 3: Testing Parameters with Poisson & Binomial Models

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7 A company claims its weekly defect rate is $\lambda = 2$. In a given week, 6 defects are observed. Test $H_0 : \lambda = 2$ against $H_1 : \lambda > 2$ by calculating $P(X \geq 6)$ under H_0.

a. 0.0166

b. 0.0527

c. 0.0345

d. 0.0045

8 At the 5% significance level, what is the conclusion for the defect test in the previous question ($p = 0.0166$)?

a. Reject H_0; the defect rate has significantly increased

b. Fail to reject H_0

c. Accept H_0

d. Change the significance level to 1%

9 A seed packet claims a 90% germination rate. A gardener tests 20 seeds and finds 15 germinate. To test $H_0 : p = 0.9$ against $H_1 : p < 0.9$, which probability must be computed?

a. $P(X \leq 15)$ where $X \sim B(20, 0.9)$

b. $P(X = 15)$ where $X \sim B(20, 0.9)$

c. $P(X \geq 15)$ where $X \sim B(20, 0.9)$

d. $P(X \leq 15)$ where $X \sim N(18, 1.8)$

Section 4: Critical Regions & Power of a Test

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10 For a test of $H_0 : \mu = 50$ vs $H_1 : \mu < 50$ using $n = 25, \sigma = 10$, find the critical value for $\bar{x}$ at the 5% level ($z_c = -1.645$).

a. 46.71

b. 45.00

c. 47.50

d. 46.00

11 In the previous test (critical region $\bar{X} < 46.71$), what is the probability of a Type II error if the true mean is $\mu = 45$?

a. 0.1964

b. 0.8036

c. 0.0500

d. 0.3085

12 What is the statistical power of the test in the previous question (where $\beta = 0.1964$)?

a. 0.8036

b. 0.1964

c. 0.9500

d. 0.0500

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